6 A hot air balloon is rising vertically at a uniform velocity of 2.0 m s^{-1}. At a height of 10 m above the ground, a ball is thrown out of the hot air balloon horizontally with a velocity of 5.0 m s^{-1}. Assume air resistance is negligible.

What is the horizontal distance travelled by the ball before hitting the ground?

A 6.2 m **B** 7.1 m **C** 7.2 m **D** 8.2 m